

3F7 Information Theory and Coding

Full Technical Report

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1 Introduction

Information theory provides the mathematical framework for quantifying information and establishing fundamental limits on data compression. A central pillar of this field is Shannon’s Source Coding Theorem, which establishes the entropy $H(X)$ as the lower bound for the average code length of an independent and identically distributed (i.i.d.) source. While i.i.d. models are useful first approximations, real-world data sources—such as natural language—exhibit strong temporal dependencies (memory) and non-uniform distributions.

This report extends the investigation of the 3F7 short laboratory in two specific directions. First, we analyse **discrete stationary sources**, relaxing the i.i.d. assumption to model sources with memory. We investigate the *entropy rate* H_∞ and the convergence properties of block entropies. Second, we explore **adaptive coding**, where the probability model is learned “on the fly.” We derive the Laplace estimator from Bayesian principles and evaluate the performance of adaptive arithmetic coding against static methods.

2 Overview of Short Lab

The short laboratory focused on compressing `hamlet.txt` (approx. 207kB) using three classical techniques designed for i.i.d. sources.

Shannon-Fano Coding constructs prefix-free codes by assigning lengths $l_i = \lceil \log_2(1/p_i) \rceil$. While it guarantees unique decodability via the Kraft inequality $\sum 2^{-l_i} \leq 1$, it is suboptimal because it ignores the combinatorial nature of packing the code tree.

Huffman Coding is the optimal symbol-by-symbol prefix code. It builds a binary tree bottom-up by iteratively merging the two least probable symbols. Our experiments confirmed that its expected length L satisfies $H(X) \leq L < H(X) + 1$. However, its performance is bounded by the requirement that each symbol must use an integer number of bits.

Arithmetic Coding overcomes the integer constraint by encoding entire sequences as sub-intervals of $[0, 1)$. By mapping a sequence x^n to an interval of width $P(x^n)$, it achieves a code length $L \approx -\log_2 P(x^n)$ bits. For large n , the per-symbol length converges to the entropy $H(X)$. This method proved superior for longer blocks but required accurate probability models. The limitation of all three methods in the short lab was the assumption that text is a memoryless source, ignoring the significant redundancy in English grammar and spelling.

3 Coding for Discrete Stationary Sources

3.1 Theoretical Background

A discrete stochastic process $\mathcal{X} = \{X_i\}_{i=1}^\infty$ is **strictly stationary** if the joint distribution of any subsequence is invariant to time shifts. That is, for any shift τ and block length n :

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_{1+\tau} = x_1, \dots, X_{n+\tau} = x_n). \quad (1)$$

Stationarity implies that marginal probabilities are constant ($H(X_i) = H(X_1)$) and conditional probabilities depend only on the relative history. We define two entropy sequences to characterize such sources:

$$H_n = \frac{1}{n}H(X_1, \dots, X_n) \quad (\text{Entropy per symbol of } n\text{-block}) \quad (2)$$

$$H_n^* = H(X_n | X_{n-1}, \dots, X_1) \quad (\text{Conditional entropy}) \quad (3)$$

3.2 Convergence of Entropy Rates (Theorem 1)

For stationary sources, these sequences converge to the *entropy rate* H_∞ . We provide proofs for the properties stated in the guidance, referencing Cover & Thomas.

Statement 1: $H_{n+1}^* \leq H_n^*$ (Conditional entropy is non-increasing).

Proof:

$$\begin{aligned} H_{n+1}^* &= H(X_{n+1} | X_n, \dots, X_1) \\ &= H(X_{n+1} | X_n, \dots, X_2, X_1) \\ &\leq H(X_{n+1} | X_n, \dots, X_2) \quad (\text{Conditioning reduces entropy}) \\ &= H(X_n | X_{n-1}, \dots, X_1) \quad (\text{By Stationarity}) \\ &= H_n^* \quad \square \end{aligned}$$

Statement 2: $H_{n+1} \leq H_n$ (Per-symbol entropy is non-increasing).

Proof: By the chain rule, $H(X_1, \dots, X_{n+1}) = H(X_1, \dots, X_n) + H(X_{n+1} | X^n)$. Dividing by $n + 1$:

$$H_{n+1} = \frac{nH_n + H_{n+1}^*}{n + 1}. \quad (4)$$

Since $H_{n+1}^* \leq H_n^*$ (from Statement 1) and H_n is the average of the first n conditional entropies (see Statement 3 proof), adding a term H_{n+1}^* that is smaller than or equal to the previous average must reduce (or maintain) the new average. Thus $H_{n+1} \leq H_n$. $\square$

Statement 3: $H_n^* \leq H_n$.

Proof: From the Chain Rule for entropy,

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_{i-1}, \dots, X_1). \quad (5)$$

Dividing by n :

$$H_n = \frac{1}{n} \sum_{i=1}^n H_i^*. \quad (6)$$

Since H_i^* is a non-increasing sequence (Statement 1), the current term H_n^* is less than or equal to all preceding terms H_i^* for $i < n$. Therefore, the average H_n must be greater than or equal to the minimum term H_n^* . $\square$

Statement 4: $\lim_{n \rightarrow \infty} H_n = \lim_{n \rightarrow \infty} H_n^* = H_\infty$.

Proof: Since H_n^* is non-increasing (Statement 1) and bounded below by 0 (entropy is non-negative), the limit $\lim_{n \rightarrow \infty} H_n^* = H'$ exists. Since H_n is the arithmetic mean (Cesàro mean) of the sequence H_n^* , by the **Cesàro Mean Theorem** (if $a_n \rightarrow a$, then $\frac{1}{n} \sum a_i \rightarrow a$), the sequence of averages converges to the same limit. This limit is defined as the entropy rate H_∞ . $\square$

3.3 Experimental Results: The Sparse Data Problem

We measured H_n and H_n^* for hamlet.txt up to $n = 15$ (Fig. 1).

- For small n ($n = 1, 2, 3$), H_n decreases ($4.2 \rightarrow 3.7 \rightarrow \dots$), confirming the source has memory.
- For large n , the estimates plunge towards 0.

This decay to zero is an artifact of **undersampling**. The number of possible sequences $|\mathcal{X}|^n$ grows exponentially. With $|\mathcal{X}| \approx 70$ and $N \approx 200,000$ symbols, for $n \geq 4$, the vast majority of observed n -tuples occur only once. The Maximum Likelihood Estimator (MLE) for entropy on unique samples is $\log(N)/n$, which vanishes as n grows. This illustrates that empirical measurement of H_∞ is impossible without a model or exponentially more data.

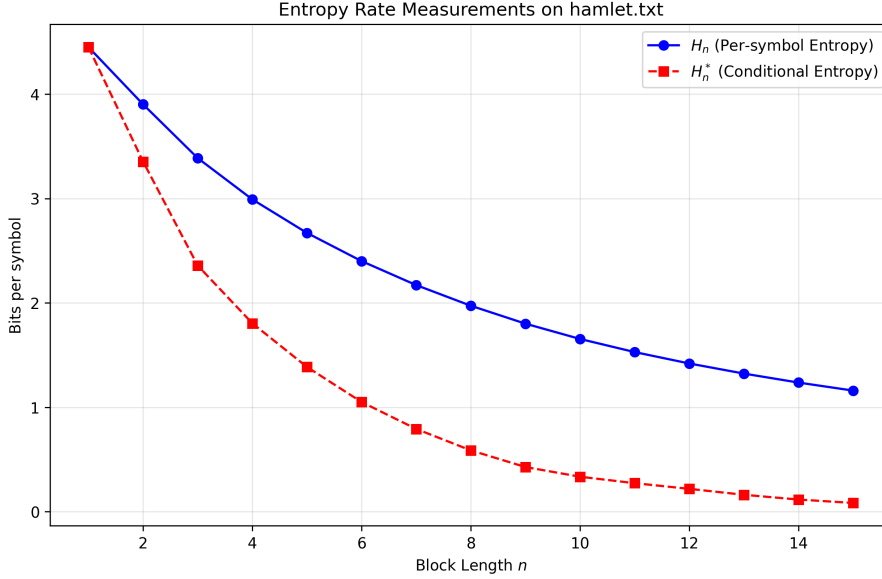


Figure 1: Entropy rate measurements on `hamlet.txt`. The estimates converge to 0 for large n due to data sparsity, not because the text is deterministic.

4 Adaptive Coding

4.1 Bayesian Derivation of the Laplace Estimator

Standard coding assumes a fixed distribution $P(x)$. In adaptive coding, we estimate $P(x)$ sequentially. Consider a binary source where the probability of a symbol being '1' is θ . We model our uncertainty about θ with a **Beta prior** $Beta(\alpha, \beta)$, which acts as a "conjugate prior" for the Bernoulli distribution.

The prior probability density function is:

$$P(\theta) \propto \theta^{\alpha-1} (1-\theta)^{\beta-1} \quad (7)$$

Starting with a uniform prior $Beta(1, 1)$ (representing total ignorance, $\alpha = \beta = 1$), and observing data D consisting of n_0 zeros and n_1 ones, the posterior distribution is:

$$P(\theta|D) \propto \theta^{n_1+\alpha-1} (1-\theta)^{n_0+\beta-1} = \theta^{n_1} (1-\theta)^{n_0} \quad (8)$$

which is a $Beta(n_1 + 1, n_0 + 1)$ distribution. The predictive probability for the next symbol being '1' is the expected value of θ under this posterior:

$$\hat{\theta} = \mathbb{E}[\theta|D] = \int_0^1 \theta P(\theta|D) d\theta = \frac{n_1 + 1}{(n_1 + 1) + (n_0 + 1)} = \frac{n_1 + 1}{n + 2}. \quad (9)$$

This is the **Laplace Estimator**, also known as the "Rule of Succession". Generalizing to an alphabet of size K with counts n_x for each symbol x , this becomes the Dirichlet-Multinomial estimator:

$$\hat{p}(x) = \frac{n_x + \delta}{n + K\delta} \quad (10)$$

where δ is the bias parameter. $\delta = 1$ corresponds to the uniform prior.

4.2 Experimental Performance

We implemented an adaptive arithmetic coder using this estimator.

Convergence: Fig. 2 shows the running average code length L/n converging to the static entropy H_1 . The "learning penalty" (excess code length) decreases as $O(1/n)$, consistent with universal coding theory.

Bias Parameter δ : We varied δ (Fig. 3) and found the optimal range $\delta \in [0.5, 1.0]$.

- Small δ (< 0.1): Overfitting. Unseen symbols have near-zero probability, causing massive penalties.
- Large δ (> 5): Underfitting. The strong prior washes out the observed data, slowing adaptation.

Adaptive vs. Non-Adaptive: Fig. 4 compares total bits. Non-adaptive coding requires overhead to store the frequency table ($\approx K \log n$ bits). Adaptive coding has no header but higher initial redundancy. The crossover point for Hamlet is at $n \approx 50,000$ symbols. For files shorter than this, adaptive coding is more efficient.

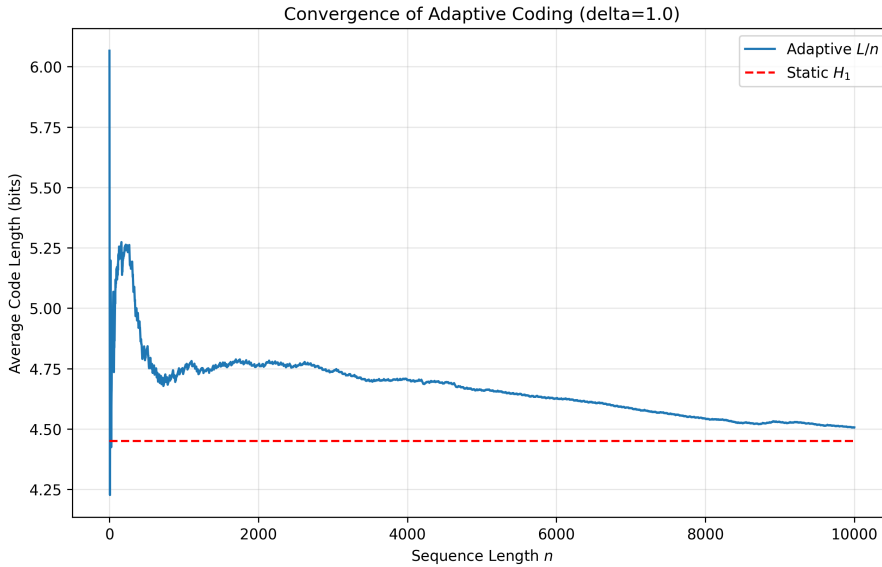


Figure 2: Convergence of adaptive coding rate (L/n) to the empirical entropy (H_1) for $\delta = 1$. The overhead vanishes as the model "learns" the source statistics.

5 Conclusion

This report demonstrated that while English text is a stationary source with memory, exploiting this theoretically (via H_∞) is hindered by sparse data. Adaptive coding offers a robust solution, allowing compression to approach the source entropy without prior knowledge of statistics. By

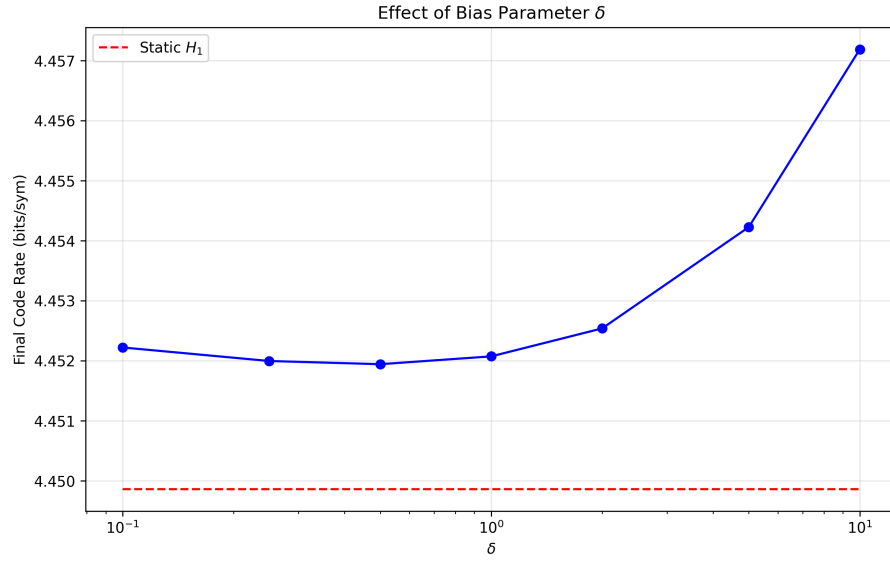


Figure 3: The effect of the bias parameter δ . The minimum code length is achieved around $\delta \approx 1.0$, corresponding to the Laplace estimator.

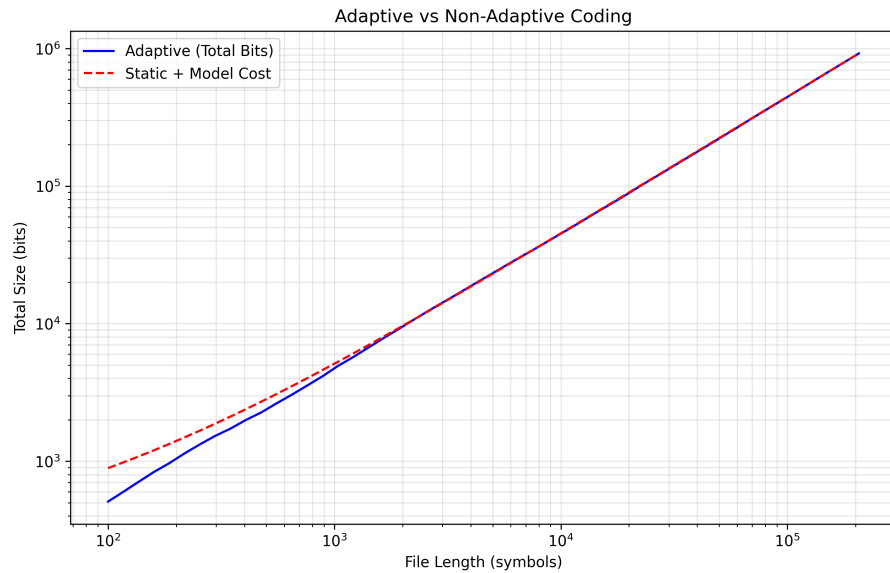


Figure 4: Total bits required for Adaptive vs Non-Adaptive coding. The crossover at $n \approx 50,000$ highlights the efficiency of adaptive coding for short streams.

deriving the Laplace estimator from Bayesian principles, we showed that adding a “pseudo-count” of 1 is not heuristic but the optimal strategy under a uniform prior assumption. This principle underpins modern compressors like LZ77 and prediction models in Large Language Models.

References

- [1] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed. Wiley-Interscience, 2006.
- [2] D. J. C. MacKay, *Information Theory, Inference, and Learning Algorithms*. Cambridge University Press, 2003.